

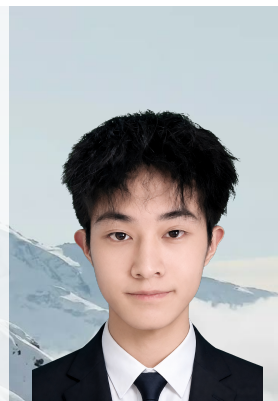
Yongxue Xu (Jerry)

Undergraduate researcher at Sun Yat-sen University

Video generation • World models • 4D scene understanding

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[Google Scholar](#) • [LinkedIn](#)



Profile

Undergraduate researcher working on video understanding, object-centric temporal modeling, and 4D dynamic reasoning. Seeking an industry research internship in multimodal foundation models, video generation, or world models.

Available from October 2026 for 6+ months, through graduation in June 2027.

Education & Toolkit

Sun Yat-sen University

SEP 2023 -- JUN 2027

B.Eng. in Intelligent Science and Technology

GPA 3.91 / 4.0 • Major GPA 3.94 / 4.0

Project 985 / 211 / Double First-Class

RESEARCH

Video understanding/generation, multimodal foundation models, world models, 4D scenes, WAM.

TOOLS & LANGUAGES

Python, C/C++, MATLAB, PyTorch; Chinese (native), English (CET-6).

Research Experience

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

Co-first author

NeurIPS 2026 submission

Developed [LMM-Track4D](#) for multi-view 4D reasoning with geometry-aware encoding, persistent state memory, and trajectory decoding. Co-developed Track4D-Bench with 526 dialogues, 23.5k frames, and 7.5k object annotations.

Achieved substantial gains over the base model on external benchmarks, particularly in dynamics reasoning, alongside 61.9% trajectory-step accuracy within 0.5 m on Track4D-Bench.

Contribution: Led model design and main experiments; co-developed the benchmark.

Learning to Reason with Persistent Object States for Video Instance Segmentation

Co-first author

ICLR 2027 submission

Developed [POSReasoner](#) for long-horizon identity reasoning, coupling persistent object memory with verified state transitions to counter identity drift and memory corruption under occlusion and reappearance.

Enhanced object recovery on LVOS v1: +13.1 $\mathcal{J}$ after 11–30 absent annotated frames and +3.5 J&F overall, without fine-tuning SAM3.

Contribution: Led method and architecture design, implementation, experiments, and manuscript preparation.

Graph Domain Adaptation Does Not End with Representation Learning

Co-first author

ICLR 2027 submission

Developed [EviGDA](#) for unsupervised graph domain adaptation: transferring node classifiers from labeled source graphs to unlabeled target graphs by fusing graph-aware and feature-only predictions.

Surpassed existing state-of-the-art methods in overall graph domain adaptation performance.

Contribution: Designed the core method and led implementation, evaluation, and manuscript development.

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach

First author / TNSE

under review

Developed [DRCIM-ML](#) with diffusion-aware role alignment and cross-layer evolutionary search, accounting for intra-layer competition, cross-layer collaboration, and node-removal robustness.

Extended RCIM from single-layer to multilayer networks, with a new formulation and a dedicated solver.

Contribution: Led problem formulation, method design, experiments, and writing; one patent pending.

Selected Publications

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue 2026

Chaoyue Li*, Yongxue Xu*, Jie Feng*, and Jiayu Ding[†]

Submitted to NeurIPS 2026; [arXiv:2605.19390](#)

Learning to Reason with Persistent Object States for Video Instance Segmentation 2026

Yongxue Xu*, Boxue Yang*, Ziqian Liu, Shaoqiu Zhang, and Linfeng Zhang[†]

Submitted to ICLR 2027

Graph Domain Adaptation Does Not End with Representation Learning 2026

Yongxue Xu*, Ziqian Liu*, Enze Zhang, and Maolin Wang[†]

Submitted to ICLR 2027

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach 2026

Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang[†]

Under review at IEEE Transactions on Network Science and Engineering

Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks 2025

Zhonghan Liu, Wenshuo Liu, Yongxue Xu, Ziqian Liu, Yichen Liu, and Shuai Wang[†]

GBCESC 2025 / Best Paper Award

* Equal contribution. [†] Corresponding author.

Research Initiatives & Collaborations

Collaborate with researchers at **HKUST**, **SJTU**, and **Westlake University** on video generation and multimodal spatiotemporal understanding. Together with peers in **InkMind.AI**, explore multimodal understanding and generation and refine AI-assisted research workflows through hands-on projects.

Additional Research

Urban road-network analysis (GBCESC 2025, Best Paper): contributed dataset construction, GNN/DQN baseline reproduction, and experimental visualization for critical-node identification.

Awards & Competitions

Sun Yat-sen University First-Class Scholarship 2023–2026

Chinese Mathematics Competition (CMC), Guangdong Region – Second Prize Dec. 2024

Mathematical Contest in Modeling (MCM), USA – Honorable Mention Feb. 2025

National Undergraduate Smart Car Competition, South China Region – Second Prize Aug. 2025

Beyond Research

Table tennis, music, and nature travel.